

ADR-007: Adopt Unity Catalog and Retire the Hive Metastore

Title: ADR-007: Adopt Unity Catalog and Retire the Hive Metastore

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Owner: Architecture Review Board, Lumenvale Technologies Inc.

Status: Accepted (implementation completed 2026-03-31)

Context

Each Lumenvale Databricks workspace had its own Hive metastore. Permissions were defined per workspace using legacy table access control, lineage was not captured, and the same table could have different permissions in different workspaces. An audit in June 2025 found 312 tables with permissions granted directly to individual users rather than groups.

Decision

We will adopt Unity Catalog as the only metastore for all Databricks workloads, with one metastore per Azure region. All tables will be migrated to Unity Catalog catalogs organized by environment and medallion layer. The Hive metastore will be retired according to the timeline below.

Timeline

- 2025-09-01: All new tables created in Unity Catalog
- 2025-10-01: Hive metastore made read-only
- 2026-01-31: All production jobs migrated
- 2026-03-31: Hive metastore decommissioned

Consequences

- Centralized, group-based access control and automatic lineage across workspaces.
- Migration effort estimated at 14 engineer-weeks, using the SYNC command for external tables and deep clone for managed tables.
- Jobs must run on Unity Catalog-enabled access modes (standard or dedicated); clusters without an access mode set are no longer supported.
- Direct user-level grants are removed; access is granted to groups only.